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Research article

Inception convolutional vision transformers for plant disease identification

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ABSTRACT

Plant disease has a considerable influence on the safety of grain output and quality. Therefore, it is crucial to detect and diagnose plant diseases. Most plant diseases are reflected in plant leaves, and accurate identification of plant diseases require specialized knowledge. Generally, it is difficult for farmers to accurately identify plant disease. Consequently, accurate and timely automatic recognition of plant disease is extremely needed in the area of precision agricultural. To solving this challenge, many classical computer vision and deep learning-based research models have been proposed. Due to the successful performance, deep learning became the preferred approach for plant disease identification. We design a novel transformer block by use of transformer architecture to model long-range features, and of soft split token embedding to capture local information from surrounding pixels and patches, in this paper. Furthermore, inception architecture and cross channel feature learning can improve the information richness, which is especially beneficial to fine-grained feature learning. The proposed model obtains higher accuracy than previous convolution and vision transformer-based models, achieves 99.94% accuracy on VillagePlant, 99.22% accuracy on ibean, 86.89% accuracy on AI2018, and 77.54% accuracy on PlantDoc. The experiment results show its preponderance over the existing models.

1. Introduction

Plant disease is considered as one of the important reasons which affect food yield and quality. It is essential to increase the production and quality of food to fulfill the demands of the rapidly growing global population. Rapid and accurate plant disease detection not only can ensure the quality and quantity of crops, but also increase economic income. Thus, it is important to detect plant diseases at the earliest stages, which deeply impact on the yield of the crops [1,2]. Rice, corn, and tomato are the essential crops in many countries like China [3]. Due to the need of a certain level of expertise and continuous observation, the manual identification of plant disease is not easy. The ratio of false plant disease recognition is slightly high for farmers to fulfill [4,5]. Therefore, it is necessary to develop automated recognition methods, accurate, timely, and economically monitor the health of crops providing helpful knowledge to the decision [6]. However, this is a challenging and very time-consuming assignment.

Recently, traditional machine learning algorithms have achieved a number of results in plant disease identification [7,8]. Saleem et al. [7] proposed a five-step method to detect plant disease type through leaf images. Johannes et al. [8] proposed statistic inference and candidate hot-spot detection algorithms for tackle identification in wild condition. Shi et al. [9] proposed a two-Dimensionality Subspace Learning Dimensionality Reduction method (2DSLDR) to apple disease recognition. Generally, the

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performance of traditional machine learning methods mainly relies on the quality of the feature extraction. But handcrafted features extraction needs abundant professional knowledge, which is subjective. Meanwhile, it is difficult to identify which information is robust and beneficial to recognition tasks from the mass of extracted features. In addition, the method is easily interfered with by noise, complex background, and the final recognition accuracy is not high.

More recently, deep learning methods, particularly Convolutional Neural Networks (CNNs), have obtained excellent achievement in computer vision, such as image recognition, face detection, face classification, and target tracking. Inspired by image classification successes, many works try CNN-based architectures for plant disease identification. Saeed et al. [1] used Visual Geometry Group (VGG) [10] model as a feature extractor. Then, partial least squares regression is used for feature selection. Lastly, the most discriminant features combined with the baggage tree classifier for final plant disease classification. Too et al. [11] focused on fine-tuning a series of CNNs architectures for image-based plant disease recognition, and found that DenseNets [12] contains a relatively fewer number of parameters to accomplish state-of-the-art performances. Data augmentation approaches can expand the number of samples, but it is difficult to solve the problem of sample diversity. Meantime, manually annotating a large number of samples is certainly a challenging problem. To solve this problem, Barbedo et al. [13] selected individual lesions and spots take the place of the whole crop leaf for the identification. How to accurately extract lesions and spots is still a difficult problem. Indeed, individual lesion and spot captured include an extensive variety of backgrounds and a wide variety of symptom features. Wang et al. [14] selected SE-ResNet-101 pre-trained on the ImageNet dataset [15] as a feature extractor, and removed the final fully connected layer, and designed a dual-stream hierarchical bilinear pooling model for plant disease identification.

Inspired by the effort of Transformer [16] in Natural Language Processing (NLP), Transformer architecture is introduced in image classification by many researchers. The Vision Transformer (ViT) [17] is the first Transformer-based image processing method. To deal with $2D$ images, the image is reshaped into a series of discrete nonoverlapping 16×16 patches. Moreover, the $2D$ patches are flattened into $1D$ tokens, and projected to D dimensions through a linear projection. Lastly, feed the linear projection of flattened patches summed with position embedding to standard Transformer encoder layers. The ViT/16 model contains about $49M$ parameters in total, which with relatively high computations and memory cost. Though the success of ViT for vision tasks at larger scale dataset is proven, the recognition accuracy is slightly below that of similar-sized convolutional neural network counterparts (e.g., GoogLeNet, EfficientNet) when trained from scratch on a midsize dataset. We hypothesize that there are two main reasons for ViT. One possible reason is that image local structures such as edges, lines and their spatial relations are ignored when splitting input images into nonoverlapping patches. However, these are important for vision tasks. Correspondingly, convolutional neural networks are good at modeling these local features by using highly correlated spatial subsampling, local receptive fields, and shared weights [18]. The second reason is the attention backbone of ViT is not efficient to CNNs. Especially, when training samples are insufficient, ViT model learns limited features richness [19]. In addition, ViT uses a bigger patch size of 16×16, and its computational and memory costs are also fairly high. Wu et al. [20] proposed Convolutional vision Transformer (CvT), which succeeds introduce convolution to the ViT model to improve performance. To solve the problem of single-scale of the output feature maps, Wang et al. [21] designed a pure Transformer backbone, and uses a pyramid architecture to decrease the feature dimension of Transformer, which can significantly reduce the computational cost. Although the pyramid architecture is beneficial to dense prediction tasks like semantic segmentation and objection detection, the performance of image classification brings little improvements.

Transformers have been certified remarkable abilities when introduced to vision tasks, but their performances are still inferior to similarly sized CNNs counterparts, e.g., ResNet [22]. In terms of plant disease identification, the performance based on the ViT architecture is also inferior to CNN-based model. We believe that there may be the following three reasons. Firstly, in order to take sequences as input of the transformer networks, images are split into sequence of patches. This results in the model only focusing on local information and ignoring the global features of the image target. Secondly, due to the fixed patch size, it is difficult to extract low-resolution and multi-scale feature maps explicitly. Lastly, the features learned by backbone of ViT are not as rich and effective as the CNN-based model when training data are not enough.

Then, we are motivated to design novel vision model to solve above problems for plant disease identification. In consideration of the state-of-the-art approaches for plant disease identification are use convolutional network networks. Meantime, inspired by the effort of Vision Transformer in image processing, we present a new network deep neural network, named Inception Convolutional Vision Transformer (ICVT), which introduces an inception convolution framework into Transformer architecture that is inherently effective and efficient, both in terms of network parameters and classification accuracy. The ICVT not only models the local spatial features of surrounding tokens, but also focuses on the learning of high-level information for plant disease identification. Specifically, the arrangement of the ICVT is illustrated in Fig. 1.

Our main contributions are as follows:

(1) We propose a novel end-to-end deep learning model, named Inception Convolutional Vision Transformer (ICVT), which can explore multi-head attention module to focus on discriminative diseased parts for plant disease identification. Meantime, soft split token is embedded to model local information from surrounding pixels and patches. Therefore, the ICVT architecture not only models the local spatial features of surrounding tokens, but also focuses on the learning of high-level information for plant disease identification.

(2) We show that the inception architecture with a vision transformer can improve the information richness and reduce redundancy in the field of plant disease identification. Cross channel feature learning by using 1×1 convolution is beneficial to receptive fields tuned to different orientation, which is especially beneficial to fine-grained feature learning for plant disease identification.

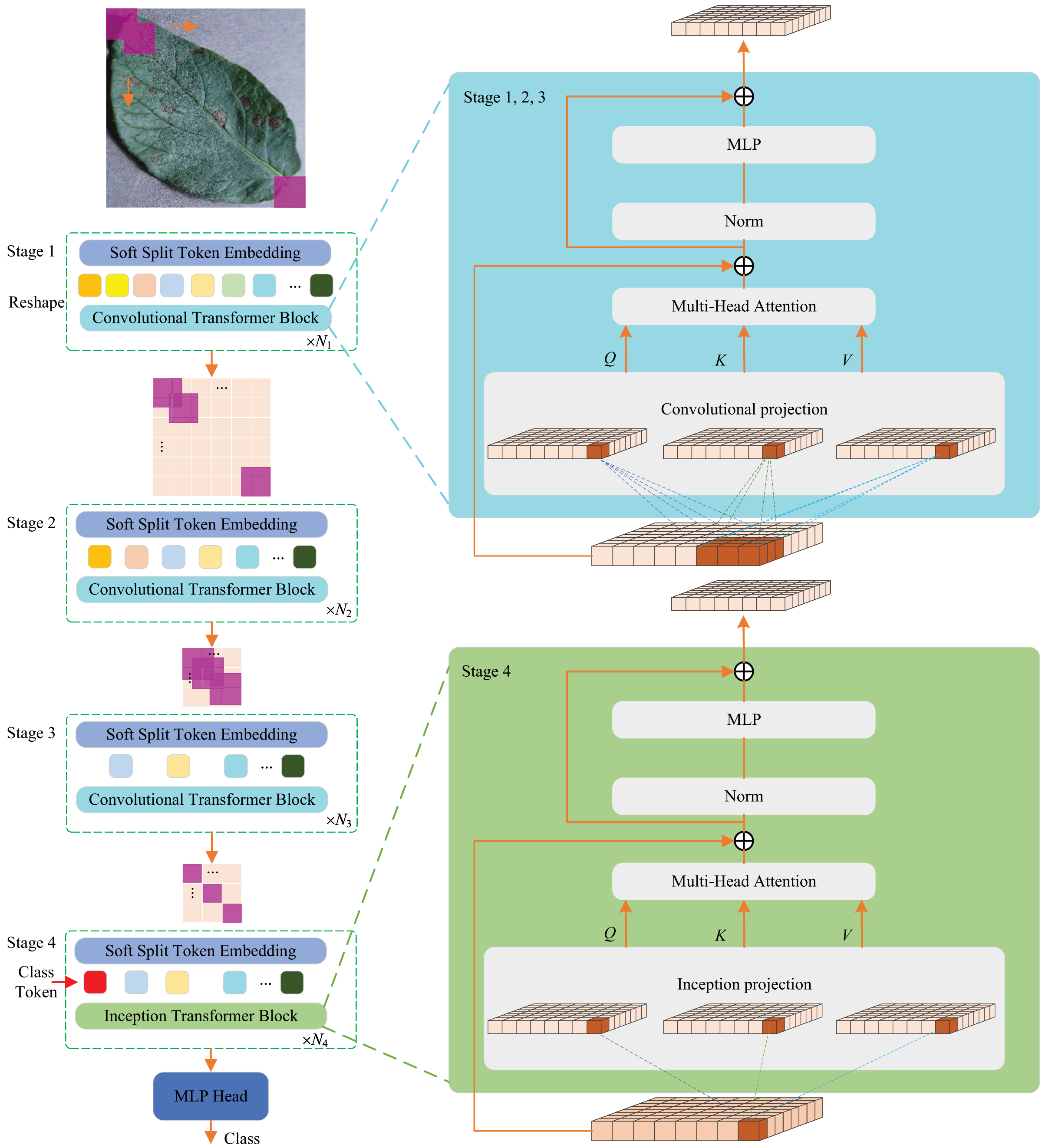


Fig. 1. The overall architecture of the ICVT method.

(3) For the plant disease identification on small-scale real scene datasets, we use the transfer learning method to avoid the overfitting problem.

The rest of this paper is organized as follows. Section 2 reviews the related works. Section 3 contains the material and methods. Section 4 focuses on the method experiments, and the experimental results are evaluated as same as contrastive analysis. Lastly, the conclusion is showed in Section 5.

2. Related works

It is critical to rapid and accurate plant disease identification for increase the yield of the crops. Majority of traditional machine learning method depend on handcrafted feature extraction that is time-consuming, complicated [23,24]. Currently, there have been more investigates to make use of deep learning, especially convolutional neural networks for plant disease identification as its powerful feature learning capabilities [25–29].

In the field of supervised learning, the CNNs models can be treated as end-to-end solutions for classification or detection problems. In [30], Brahimi et al. utilized a CNNs architecture for tomato leaf disease detection. The AlexNet and GoogleNet pre-trained models are fine-tuned on tomato datasets. Then the models as feature extractors to extract features, and the extracted features are fed to SVM for disease classification and obtained the greatest accuracy 99.86% and 99.18% on PlantVillage dataset [21]. Similarly, Saeed et al. [1] proposed an crop disease identification architecture using VGG19 model as feature extractor, and using Partial Least Squares (PLS) regression as feature selector. Then, the PLS projection method is used to select the most discriminant features, and using baggage tree classifier to implement the final identification. DenseNet-201 architecture is used in [31] to identify plant diseases. Fusion method [32] for plant disease identification includes four core steps, infected regions detection, feature extraction, feature selection and classification. The best classification accuracy is 98.60% on the CASC-IFW. Hernandez et al. [33] proposed probabilistic programming method using Bayesian deep learning, and obtained 96% accuracy on PlantVillage. Ji et al. [34] proposed a UnitedModel architecture by using of ResNet-50 [22] and Inception V3 [35], which can effectively obtain complementary discriminative features. The UnitedModel achieved a test accuracy 98.57% on the grape disease dataset. Fang et al. [36] designed an under-clustering method to learn the most possible clusters making use of cross-iterative k-means clustering. The network is used to evaluate the similarity, and obtained 89.5%, 52.1%, and 84.9% on three separate datasets. To extract the discriminative features, Turkoglu et al. [37] proposed a hybrid neural network by combining AlexNet, ResNet-18, ResNet-50, ResNet-101, DenseNet-201, and GoogleNet for feature learning and SVM for identification. Beyond that, the authors built a Turkey-PlantDataset, which consisting of 4447 images in 15 classes. Recently, Yuan et al. [38] designed a spatial pyramid-oriented encoder–decoder cascade CNN architecture for plant disease leaf detection and segmentation. The above studies have showed the ability of CNNs in the area of plant disease recognition. It is also revealed that CNN-based plant disease identification architectures are stimulated mainly by base frameworks such as GoogleNet, ResNet, and Inception.

Currently, attention mechanisms have gained a lot of attention in various fields, including plant disease identification. Karthik et al. [39] introduced attention mechanism in a deep residual CNN for tomato leaves infection detection. Zeng et al. [40] proposed a residual CNN architecture with self-attention to extract features of crop disease spots for crop disease identification. Their model obtained 95.33%, 98.0% on AES-CD9214 and MK-D2, respectively. To expand the database and increase the diversity of sample data, Cap et al. [41] proposed a LeafGAN model, which by using GAN with an own attention mechanism for image-to-image translation. Zhou et al. [42] proposed a progressive learning network with attention block for complex background vegetables disease detection, and achieved 15% improve in identification accuracy. Chen et al. [27] introduced squeeze-and-excitation block in MobileNet v2 [43] network for detection of plant disease leaves. Similarly, Chen et al. [44] designed depth-wise separable convolution with MobileNet v2 for plant disease detection. In [45], a MobileNet [46] with multi-head self-attention model proposed for rice disease identification, and obtained identification accuracy 94.65% for rice disease.

More recently, the ViT architecture is introduced in plant disease identification [47–49]. Thakur et al. [47] proposed a PlantXViT network for plant disease detection. The PlantXViT network is consist of vision transformer blocks and CNNs architecture. Borhani et al. [48] proposed lightweight networks for real-time crop disease identification with ViT architecture. Lu et al. [50] designed the GhostNet with vision transformer blocks to plant disease identification, namely Ghost enlightened Transformer (GeT). The network obtained accuracy 98.14% on the GLDP12k dataset, which contains of 12,615 vine images in 11 classes. To extract the most effective and relevant features for olive disease classification, Alshammari et al. [49] designed a hybrid deep learning-based model, which combines the vision transformer architecture and CNN architecture.

Among above-mentioned deep learning-based methods, PlantXViT [47], GeT [50], and CvT [20] are most closely related to the ICVT. In [47], PlantXViT is built by combining two convolutional blocks of VGG16, one inception block, and ViT architecture. The GeT network is mainly based on standard ViT blocks and Ghost-convolution [51] for grape leaf disease identification in the field. In the above two papers, ViT and convolution are two independent modules, convolution is used for image preprocessing and feature extraction, and ViT module completes self-visual attention learning and final plant disease identification. In contrast, our ICVT model embeds the convolutional projection and the inception transformer into ViT to improve the information richness and learn fine-grained feature. In [20], the CvT network is used for the task of image classification, but the aim of our method is plant disease identification. Plant disease identification need to pay more attention to the subtle feature differences between leaf images. Therefore, we embed the inception transformer block in stage 4 of the ICVT model to extract fine-grained features. Meanwhile, we modified the number and size of convolution kernels in convolution projection blocks for better focus on plant disease areas.

3. Material and methods

3.1. Image datasets

In order to design a proper plant disease identification method, the designed models are trained and test on public datasets. All the images are collected from the various diseased and healthy leaves, which belonging to four different datasets.

First one is PlantVillage dataset [21], which contains 54,305 images divided into 38 plant disease categories, and belonging to 14 different plant species. Moreover, the dataset contains a class for 1143 background images without leaf. Then, the total number of images is 55,448 in the dataset. All the samples in the PlantVillage dataset are colored images of 256×256 sizes. Fig. 2 shows 8 various sample images selected randomly.

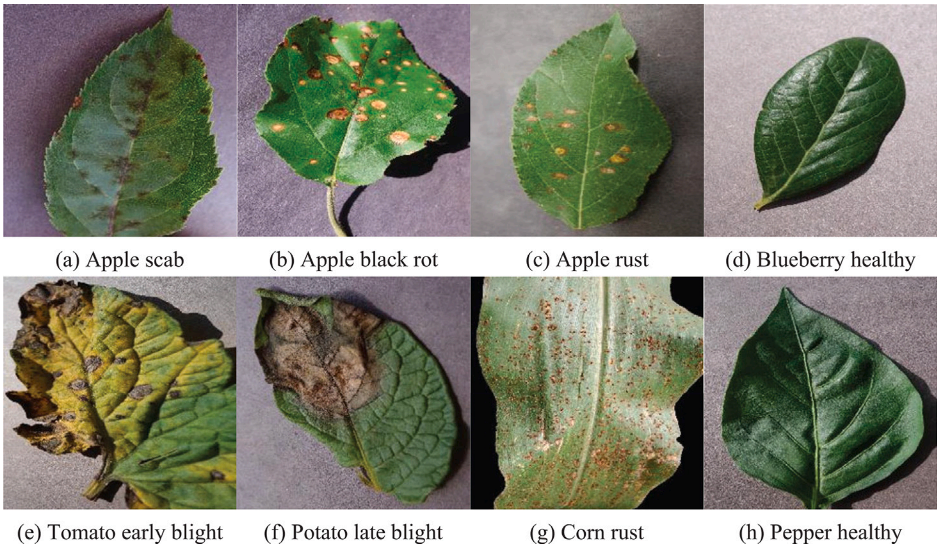


Fig. 2. Sample images of plant diseases on the PlantVillage dataset.

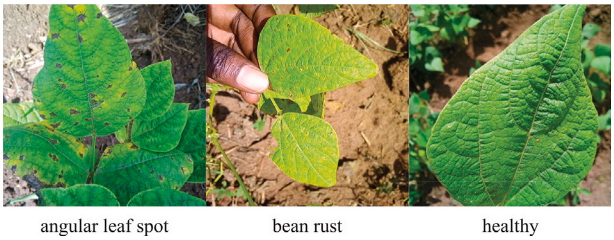


Fig. 3. Sample images of plant diseases on the ibean dataset.

Table 1
Brief of ibean dataset.

Disease	Number of images
Healthy	428
Bean rust	436
Angular leaf spot	432
Total	1296

The second dataset is ibean leaf image dataset. The sample data is taken in the field by professional. The data of leaf images are split into three classes: Angular Leaf Spot, Bean Rust diseases, and a healthy class. Table 1 shows a description of the ibean dataset. The agricultural experts participate in data collection and all sample images were collected from the field. In Fig. 3, some sampled images from the ibean are showed. All images have the same size, and the images are taken in the actual scene with a complicated background.

The third dataset is AI2018. The competition provided an image disease dataset with ten different plant categories. The dataset is divided into 61 categories by disease, degree and species. To alleviate the impact of sample imbalance problem on model performance, We remove two sample categories that only contain one image, in this paper. Lastly, the AI2018 contains 31,718 images in the training set, 4540 images for validation, and 8671 images for the test. Science the sample image in the test set does not provide labels, this paper uses the validation set data as the model performance evaluation. In addition, the dataset does not give the disease category of the specific image, and directly labels the category labels such as 0, 1, and 2. In Fig. 4, some sampled images from the AI2018 are showed.

Fig. 4. Sample images of plant diseases on the AI2018 dataset.

(a) Apple rust leaf (b) Apple Scab Leaf (c) Bell pepper leaf (d) Bell_pepper leaf spot

(e) Corn leaf blight (f) Corn rust leaf (g) Tomato leaf bacterial spot (h) Tomato leaf mosaic virus

Fig. 5. Sample images of plant diseases on the PlantDoc dataset.

The fourth dataset is PlantDoc [52]. The dataset contains 2598 images which has 13 plant categories with 27 species by disease, of which 80% are selected for the training and the remaining for the test set. Different form the PlantVillage and AI2018, the images of the PlantDoc are taken in the field by using mobile phone cameras. In Fig. 5, some sampled images from the PlantDoc are showed.

3.2. Inception Convolutional Vision Transformer (ICVT)

The overall architecture of the ICVT is showed in Fig. 1. We introduce soft split token embedding and depth-wise convolution-based works to the Vision Transformer architecture. Four stages of hierarchy design in total are applied in this work. The first three stages have soft split embedding and depth-wise convolutional transformer block. The last stage consists of soft split embedding and inception transformer block.

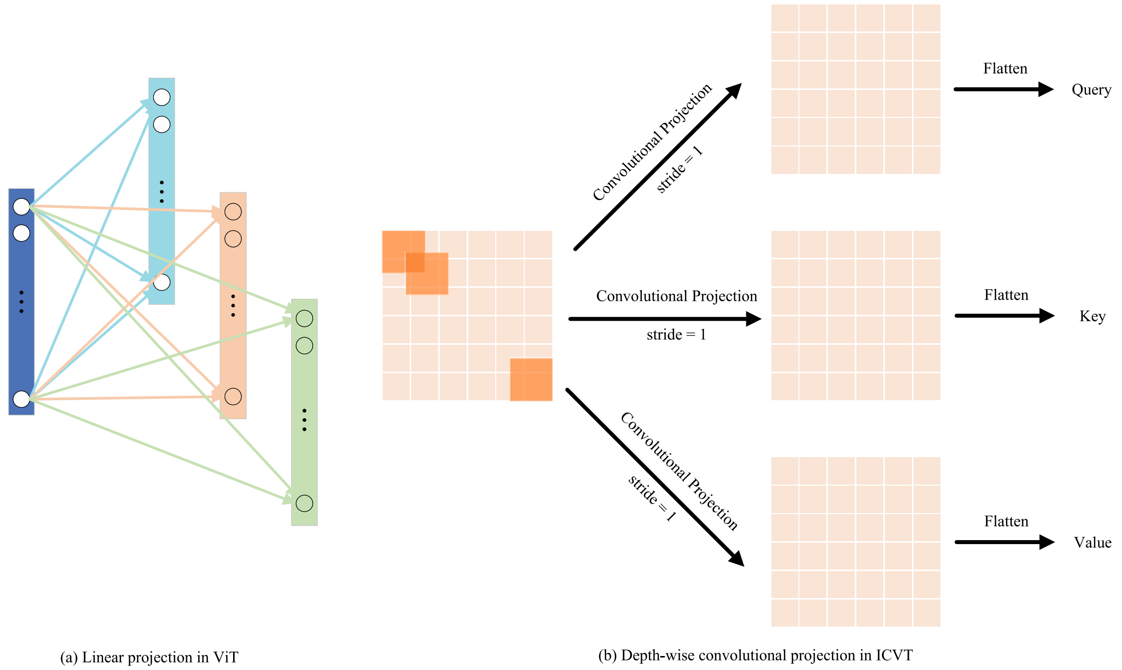


Fig. 6. Two projection architectures.

3.2.1. Soft Split Token Embedding (SSTE)

The soft split operation model local information from surrounding pixels and patches. In this work, we select convolution to implement soft split token embedding. Formally, given an input x_{i-1} of size $H_{i-1} \times W_{i-1} \times C_{i-1}$ of an image or a 2D-reshaped output map from a previous stage, we feed it to a convolutional layer to learn a function $f(.)$ with the stride of s, the kernel size of $S \times S$, and p padding, and the kernel number of C_i, where $S > s$. The size of output $f(x_{i-1}) \in \Re^{H_i \times W_i \times C_i}$ can be formulated as:

$$H_i = \lfloor \frac{H_{i-1} - S + 2P}{s} \rfloor + 1, W_i = \lfloor \frac{W_{i-1} - S + 2P}{s} \rfloor + 1 \tag{1}$$

We flatten $f(x_{i-1})$ into size $H_i \times W_i \times C_i$ in spatial dimensions, and it normalized by layer normalization [53]. By this method, the token sequence length is progressively decreased, and the token feature dimension can be increased. Similar to feature layers of CNNs, this gives the token a larger receptive field to represent increasingly complex visual patterns.

3.2.2. Depth-wise convolutional transformer block

Compared to the standard Transformer block [17], we use a depth-wise convolutional projection layer as the Transformer block to model local spatial context information in the first three stages. As showed in Fig. 6, the input tokens are reshaped into a 2D token map, firstly. Then, a depth-wise convolutional projection is used with kernel size $s{\times}s$. Finally, the mapped tokens are flattened into 1D as inputs of the multi-head attention module. This can be formulated as:

$$x_i^{q/k/v} = Flatten(Conv2D(Reshape2D(x_i), s)) \tag{2}$$

where $x_i^{q/k/v}$ is the token input for $Q/K/V$ matrices of layer i, x_i is the soft split token prior to the convolutional projection, Conv2D is a convolution with convolution kernel size $s \times s$. Compared with linear projection, the number of the depth-wise convolutional projection is about $25M$, which much lower than $49M$ parameters of the ViT/16 [17].

3.2.3. Inception transformer block

In plant disease identification, the apparent features of different types of diseases often have only subtle apparent differences, effectively learning the fine-grained features of disease images is crucial to the identification of plant diseases. Therefore, the model need pay more attention to subtle and local differences for better plant disease identification. Inspired by the success of the Inception module [53,54] in increasing the depth of the convolutional neural networks, we introduce Inception projection to the ViT network to learn more complex functions for modeling fine-grained features, namely the Inception transformer block.

Compared to CvT [20], we add stage four with Inception Transformer Block to increase the depth to pay more attention to fine-grained features and model more local spatial context information. Table 2 shows the detailed structure of ICVT model. 224 × 224 is

Table 2
Detailed setting of the ICVT.

	Output size	Layer name	ICVT
Stage 1	56×56	SSTE	7×7, 64, stride = 4
	56×56	Convolutional projection MHSA MLP	$\begin{bmatrix} 3 \times 3, 64 \\ H_1 = 1, D_1 = 64 \\ R_1 = 4 \end{bmatrix} \times 1$
Stage 2	28×28	SSTE	3×3, 192, stride = 2
	28×28	Convolutional projection MHSA MLP	$\begin{bmatrix} 3 \times 3, 192 \\ H_2 = 3, D_2 = 192 \\ R_2 = 4 \end{bmatrix} \times 2$
Stage 3	14×14	SSTE	3×3, 384, stride = 2
	14×14	Convolutional projection MHSA MLP	$\begin{bmatrix} 3 \times 3, 384 \\ H_3 = 6, D_3 = 384 \\ R_3 = 4 \end{bmatrix} \times 10$
Stage 4	14×14	SSTE	1×1, 384, stride = 1
	14×14	Inception Projection MHSA MLP	$\begin{bmatrix} 1 \times 1, 384 \\ H_4 = 6, D_4 = 384 \\ R_4 = 4 \end{bmatrix} \times 3$
Head	1×1	Linear	Class No.
Parameters			$25M$

the default size of the input image. SSTE: Soft Split Token Embedding. H_i indicates the number of heads in the ith MHAS component. D_i is the number of embedding feature dimensions. R_i is the feature dimension dilatation coefficient in the ith MLP layer.

3.2.4. ICVT with transfer learning

Our ICVT model is a very deep neural network which incorporates vision transformer blocks and convolution operation. Current, the dataset of plant disease identification is small, which cannot meet the effective training of ICVT, and is prone to overfitting. To solve this problem, we applied our ICVT model using transfer learning for plant disease identification. First, the ICVT model is pre-trained on the ImageNet dataset [15] from scratch. Then, the pre-trained ICVT model is fine-tuned on the plant disease datasets. Specifically, the number of units of the final linear layer is changed to the plant disease categories of the dataset. The weights of the ICVT have been initialized with pre-trained weights, and have been fine-tuned during the training.

4. Experimental results and discussions

The ICVT-based plant disease identification model is trained with different image samples various crops in the computer having Intel Core i5-6600K processor, 32 GB RAM, TITAN X GPU. Programming framework is Python with PyTorch, Keras. Considering the memory limitations of CPU and GPU, the batch size of training networks is taken as 32. AdamW [55] is applied as an optimizer because of its strong performance over the Adam optimizer. The weight decay is 0.05, and use a cosine learning rate decay scheduler and initial learning rate of 0.02 is adopted. By default, the input size of the images is 224 × 224. The dataset is separate into two parts, the test set and the training set. The division of training set and test set is determined according to the specific data set. In addition, the training set augmentation and regularization methods are taken in ViT [17].

4.1. Evaluation metrics

Performance metrics help in measuring the quality of the classification method, and different metrics focus on different aspects of the evaluation model. In this work, we use four evaluation methods such as precision, accuracy, recall, and F1-score to appraise the performance of the network. Accuracy is the rate of correctly classified images out of total number of samples. Precision is a fraction of the correct prediction from all positive images. True Positive (TP) is correct number of sample labels that are predicted correctly with reference to the ground truth. True Negative (TN) is negative sample labels that are predicted correctly with respect to the ground truth. False Positive (FP) is negative samples that are predicted to be positive samples. False Negative (FN) is those positive images that are predicted to be negative samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

Table 3
Performance evaluation on the PlantVillage for each class.

Plant categories	Disease categories	Class labels	Precision	Recall	F1-score
Apple	Scab	0	1.000	1.000	1.000
	Black rot	1	1.000	1.000	1.000
	Cedar apple rust	2	1.000	1.000	1.000
	Healthy	3	1.000	1.000	1.000
Background	Background	4	0.9956	1.000	0.9978
Blueberry	Healthy	5	1.000	1.000	1.000
Cherry	Powdery mildew	6	1.000	1.000	1.000
	Healthy	7	1.000	1.000	1.000
Corn	Gray leaf spot	8	0.9802	0.9706	0.9754
	Common rust	9	1.000	1.000	1.000
	Northern leaf blight	10	0.9848	0.9898	0.9873
	Healthy	11	1.000	1.000	1.000
Grape	Black rot	12	1.000	1.000	1.000
	Esca (Black Measles)	13	1.000	1.000	1.000
	Leaf blight	14	1.000	1.000	1.000
	Healthy	15	1.000	1.000	1.000
Orange	Haunglongbing	16	1.000	1.000	1.000
Peach	Bacterial spot	17	1.000	1.000	1.000
	Healthy	18	1.000	1.000	1.000
Pepper	Bell bacterial spot	19	1.000	1.000	1.000
	Bell healthy	20	1.000	1.000	1.000
Potato	Early blight	21	1.000	1.000	1.000
	Late blight	22	1.000	1.000	1.000
	Healthy	23	1.000	1.000	1.000
Raspberry	Healthy	24	1.000	1.000	1.000
Soybean	Healthy	25	1.000	1.000	1.000
Squash	Powdery mildew	26	1.000	1.000	1.000
Strawberry	Leaf scorch	27	1.000	1.000	1.000
	Healthy	28	1.000	1.000	1.000
Tomato	Bacterial spot	29	1.000	1.000	1.000
	Early blight	30	1.000	1.000	1.000
	Late blight	31	1.000	1.000	1.000
	Leaf mold	32	1.000	1.000	1.000
	Septoria leaf spot	33	0.9972	1.000	0.9986
	Spider mites	34	1.000	1.000	1.000
	Target spot	35	1.000	0.9964	0.9982
	Yellow leaf curl virus	36	1.000	1.000	1.000
	Mosaic virus	37	1.000	1.000	1.000
	Healthy	38	1.000	0.9969	0.9984
	Average		0.9989	0.9988	0.9989

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{6}$$

4.2. Performance of the experiments

The main purpose of this section is to approve the success of ICVT network in the identification of plant disease and to compare with the results of the state-of-the-art architectures. All experiments are implemented on mentioned above four datasets. The recall, precision, and F1-score for each class of the PlantVillage are showed in Table 3. The average value is 0.9989, 0.9988, 0.9989 for precision, recall, and F1-score respectively. Similarly, the results gained by ICVT are showed in Tables 4–6 for the AI2018, ibean, and PlantDoc. The confusion matrix of the ibean, PlantVillage, AI2018, and PlantDoc are showed in Figs. 7, 8, 9, and 10, respectively. Observing from the confusion matrix, the proposed ICVT method has very few false positives and false negatives on the PlantVillage and ibean datasets. Through comparative analysis of misclassified samples, we found that most of the misclassifications occurred when leaves with different diseases exhibited similar epigenetic features. As showed Figs. 11, 12, and 13, the misclassified image samples are equally easy to make false identifications even for experienced farmer.

Table 4
Performance evaluation on the AI2018 for each class.

Disease categories	Precision	Recall	F1-score
0	0.9939	0.9704	0.9820
1	0.7419	0.7667	0.7541
2	0.6111	0.8148	0.6984
3	0.8889	0.3333	0.4848
4	0.8525	0.7536	0.8000
5	0.7119	0.8235	0.7636
6	0.6875	0.7586	0.7213
7	0.8267	0.8732	0.8493
8	1.0000	1.0000	1.0000
9	0.9767	1.0000	0.9882
10	0.6667	0.8159	0.7480
11	0.8600	0.6515	0.7414
12	0.8235	0.6364	0.7179
13	0.8256	0.9595	0.8875
14	0.9375	0.7627	0.8411
15	0.5000	0.1111	0.1818
16	0.9271	0.9889	0.9570
17	1.0000	1.0000	1.0000
18	0.6741	0.8996	0.7707
19	0.8430	0.5534	0.6682
20	0.9167	0.9167	0.9167
21	0.9145	0.8770	0.8954
22	0.8991	0.8909	0.8950
23	0.9683	1.0000	0.9839
24	0.9800	1.0000	0.9899
25	0.7209	0.7750	0.7470
26	0.7818	0.7963	0.7890
27	0.9760	0.9951	0.9854
28	0.7879	0.8966	0.8387
29	0.9848	0.8904	0.9353
30	0.8205	0.8889	0.8533
31	0.8636	0.8906	0.8769
32	1.0000	1.0000	1.0000
33	0.8636	0.7037	0.7755
34	0.8696	1.0000	0.9302
35	0.9080	0.9518	0.9294
36	0.9884	0.9884	0.9884
37	0.7609	0.7609	0.7609
38	0.9203	0.9203	0.9203
39	0.8611	0.8611	0.8611
40	0.9474	0.8571	0.9000
41	0.7941	0.7105	0.7500
42	0.8922	0.9430	0.9169
43	1.0000	0.6667	0.8000
44	0.8222	0.8043	0.8132
45	0.7800	0.8125	0.7959
46	0.5000	0.6000	0.5000
47	0.3333	0.6000	0.4286
48	0.8226	0.8500	0.8361
49	0.9369	0.9043	0.9204
50	0.8442	0.8442	0.8442
51	0.7105	0.6923	0.7013
52	0.8114	0.7030	0.7533
53	0.8439	0.9037	0.8728
54	1.0000	1.0000	1.0000
55	0.9714	0.9189	0.9444
56	0.7500	1.0000	0.8571
57	1.0000	0.8333	0.9091
58	1.0000	1.0000	1.0000
Average	0.8457	0.8326	0.8300

The average values of accuracy, precision, recall and F1-score for each class on the PlantVillage dataset are given in Table 7. We can see that ICVT and CVT models offered close results, but the ICVT model is more successful than the CVT on the PlantVillage. In Table 8, we show average values of deep learning-based methods on the PlantDoc. The results show that the ICVT method can also achieve the highest accuracy and precision in practical scenarios.

Table 5
Performance evaluation on the ibean for each class.

Disease categories	Class labels	Precision	Recall	F1-score
Angular leaf spot	0	0.9773	1.000	0.9885
Bean rust	1	1.000	0.9767	0.9882
Healthy	2	1.000	1.000	1.000
Average		0.9924	0.9922	0.9922

Table 6
Performance evaluation on the PlantDoc for each class.

Disease categories	Class labels	Precision	Recall	F1-score
Apple Scab Leaf	0	0.8333	1.0000	0.9091
Apple leaf	1	0.7500	0.6667	0.7059
Apple rust leaf	2	1.0000	0.8000	0.8889
Bell_pepper leaf	3	0.7778	0.8750	0.8235
Bell_pepper leaf spot	4	0.5714	0.4444	0.5000
Blueberry leaf	5	0.9091	0.9091	0.9091
Cherry leaf	6	0.8182	0.9000	0.8571
Corn Gray leaf spot	7	0.1429	0.2500	0.1818
Corn leaf blight	8	0.6667	0.5000	0.5714
Corn rust leaf	9	0.9000	0.9000	0.9000
Peach leaf	10	1.0000	0.7778	0.8570
Potato leaf early blight	11	0.6667	0.5000	0.5714
Potato leaf late blight	12	0.8000	0.5000	0.6154
Raspberry leaf	13	0.8750	1.0000	0.9333
Soyabean leaf	14	1.0000	0.8750	0.9333
Squash Powdery mildew leaf	15	1.0000	1.0000	1.0000
Strawberry leaf	16	1.0000	1.0000	1.0000
Tomato Early blight leaf	17	0.6667	0.0000	0.7619
Tomato Septoria leaf spot	18	1.0000	0.3636	0.5333
Tomato leaf	19	0.8571	0.7500	0.8000
Tomato leaf bacterial spot	20	0.3182	0.7778	0.4516
Tomato leaf late blight	21	0.8182	0.9000	0.8571
Tomato leaf mosaic virus	22	0.8571	0.6000	0.7059
Tomato leaf yellow virus	23	0.8571	1.0000	0.9231
Tomato mold leaf	24	0.7143	0.8333	0.7692
Grape leaf	25	1.0000	1.0000	0.9600
Grape leaf black rot	26	1.0000	0.8750	0.9333
Average		0.8074	0.7407	0.7723

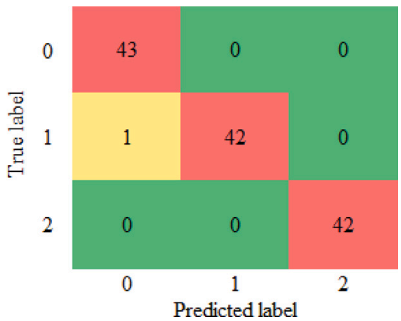


Fig. 7. Confusion matrix of ICVT for the ibean.

From Figs. 14 to 17, we show the Receiver Operating Characteristic (ROC) curves of the proposed method for PlantVillage, AI2018, ibean, and PlantDoc datasets, respectively. The Area Under the Curve (AUC) on the first three datasets is equal to 1 or 0.99. In the PlantDoc, it also achieved above 0.93. These ROC curves also manifest that the proposed method has good recognition results.

In Table 9, we show the performance comparison of the proposed ICVT with existing research works. As can be seen, compared to traditional convolutional neural networks-based method, our results outperform the previous best models by 0.86% on the AI2018,

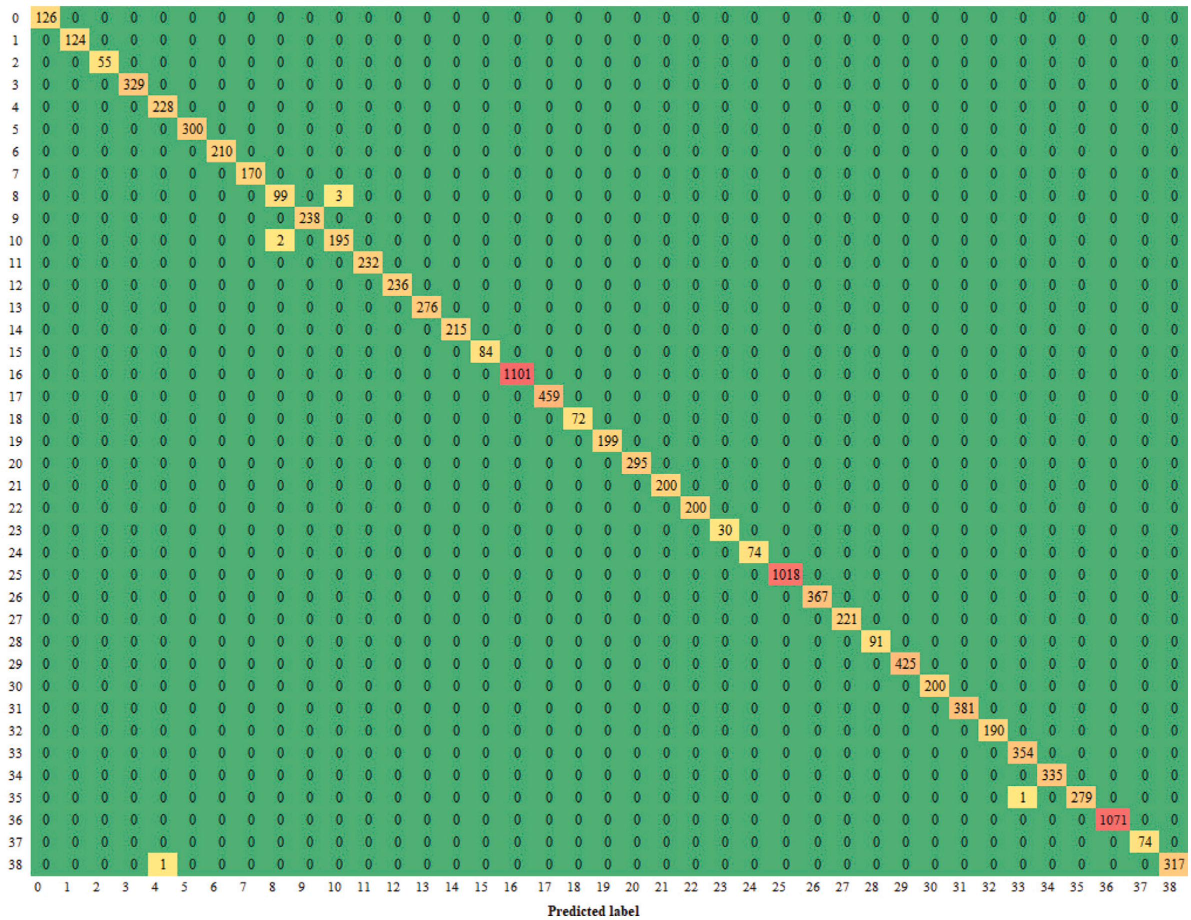


Fig. 8. Confusion matrix of ICVT for the PlantVillage.

Table 7
Average values of deep learning-based methods on the PlantVillage.

Model	Ave Acc	Ave precision	Ave recall	Ave F1-score
ICVT	**0.9994**	**0.9989**	**0.9988**	**0.9989**
ViT/16 [17]	0.9958	0.9974	0.997	0.9972
CVT [20]	0.9991	0.9990	0.9982	0.9986
EfficientNet-B0 [56]	0.9981	0.9685	0.9626	0.9655
EfficientNet-B5 [56]	0.9991	0.9842	0.9831	0.9836
EfficientNet-B7 [56]	0.9986	0.9760	0.9723	0.9742
ResNet-50 [22]	0.9977	0.9653	0.9544	0.9598
VGG-16 [10]	0.9981	0.9683	0.9626	0.9654

Table 8
Average values of deep learning-based methods on the PlantDoc.

Model	Average accuracy	Average precision
ICVT	**0.7754**	**0.7723**
DHBP [14]	0.7506	–
T-CNN (based on ResNet-101) [25]	0.7558	0.7444
VGG-16	0.5852	0.5747
Inception v3	0.6647	0.6592
ResNet-101	0.6529	0.6413

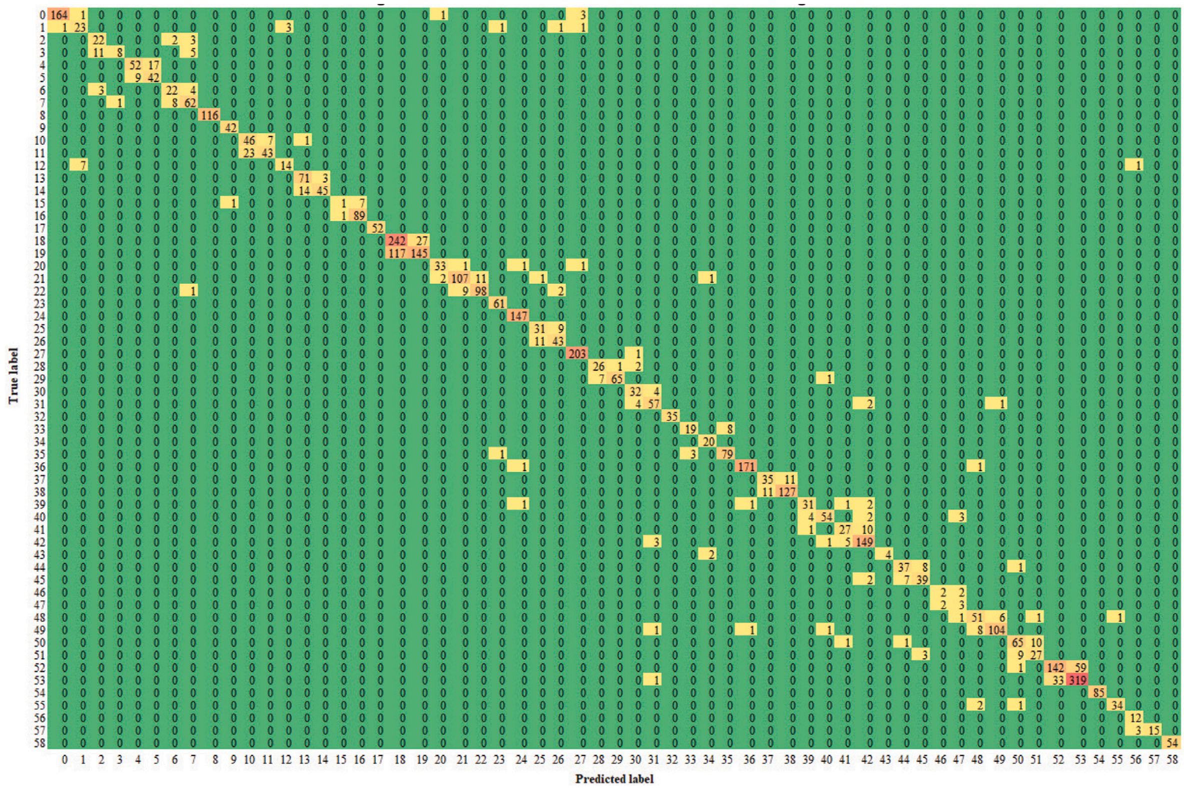


Fig. 9. Confusion matrix of ICVT for the AI2018.

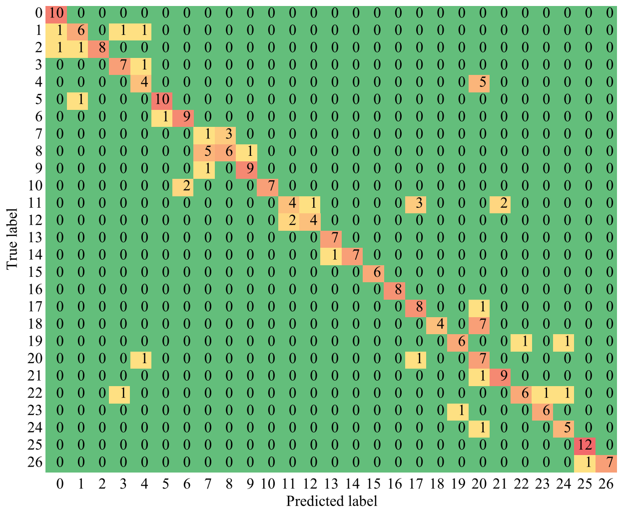
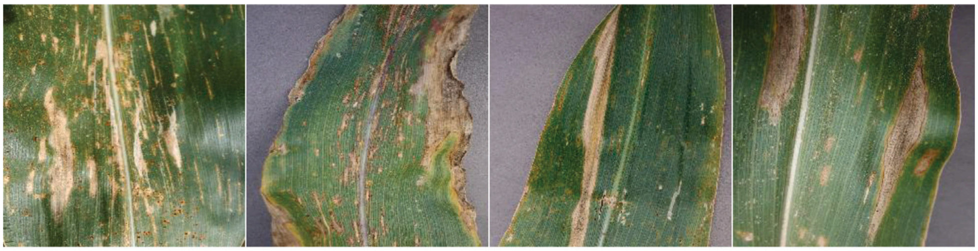


Fig. 10. Confusion matrix of ICVT for the PlantDoc.

0.03% on the VillagePlant, 1.56% on the ibean, and 14.27% on the PlantDoc. We also compared the proposed ICVT with VIT related methods such as ViT and CVT, our average accuracy outperforms CVT method 0.31% on the AI2018, 0.03% on the VillagePlant, 3.91% on the ibean, and on the 1.96% PlantDoc. The superior performance shows the discriminative capacity of the proposed



(a) Gray leaf spot (b) Northern Leaf Blight

Fig. 11. Misclassified samples on the PlantVillage.

(a) class 18 (b) class 19 (c) class 52 (d) class 53

Fig. 12. Misclassified samples on the AI2018.

(a) Tomato leaf bacterial spot (b) Tomato Septoria leaf spot (c) Tomato Early blight leaf (d) Tomato leaf late blight

Fig. 13. Misclassified samples on the PlantDoc.

Table 9
Comparison of average accuracy with the state-of-the-art methods for plant disease classification.

Model	VillagePlant	AI2018	ibean	PlantDoc
ResNet-50 [22]	99.63%	85.97%	97.66%	58.63%
VGG-16 [10]	99.81%	85.39%	96.85%	58.52%
EfficientNet-B5 [56]	99.91%	86.03%	97.50%	63.27%
EfficientNet-B7 [56]	99.86%	85.56%	97.44%	62.28%
ViT/16 [17]	99.58%	86.41%	94.53%	–
CVT [20]	99.91%	86.58%	95.31%	73.56%
DHBP [14]	99.41%[a]	–	–	75.06%[a]
T-CNN (based on ResNet-101) [25]	99.70%[a]	–	–	75.58%[a]
ICVT	**99.94%**	**86.89%**	**99.22%**	**77.54%**

[a]Indicates the experimental data of the author of the original paper.

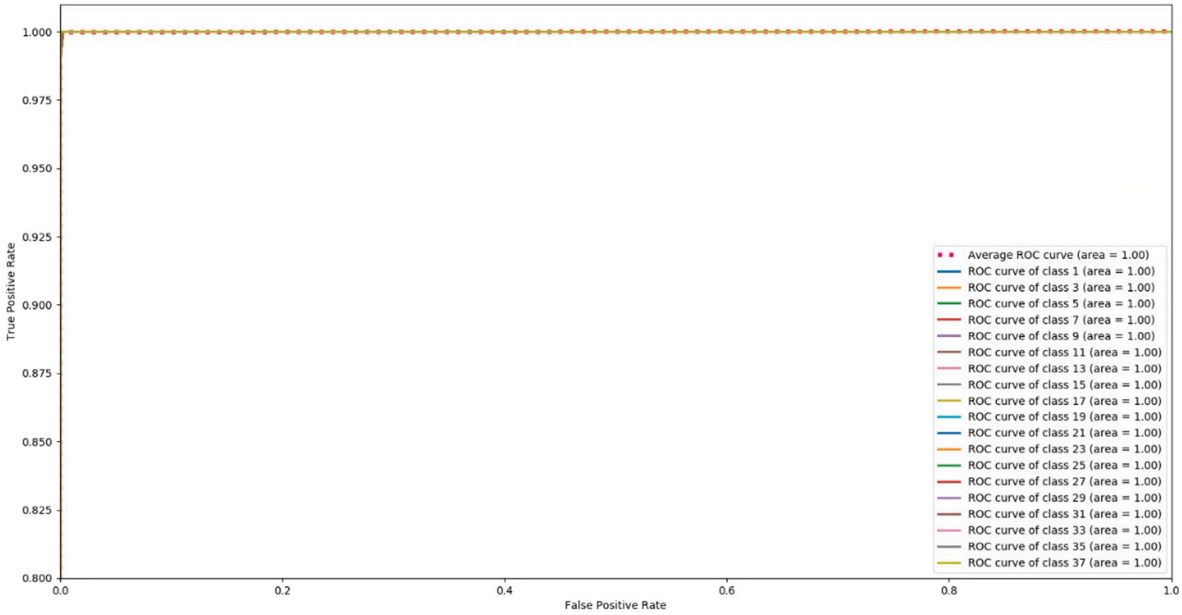


Fig. 14. ROC curves on the PlantVillage dataset.

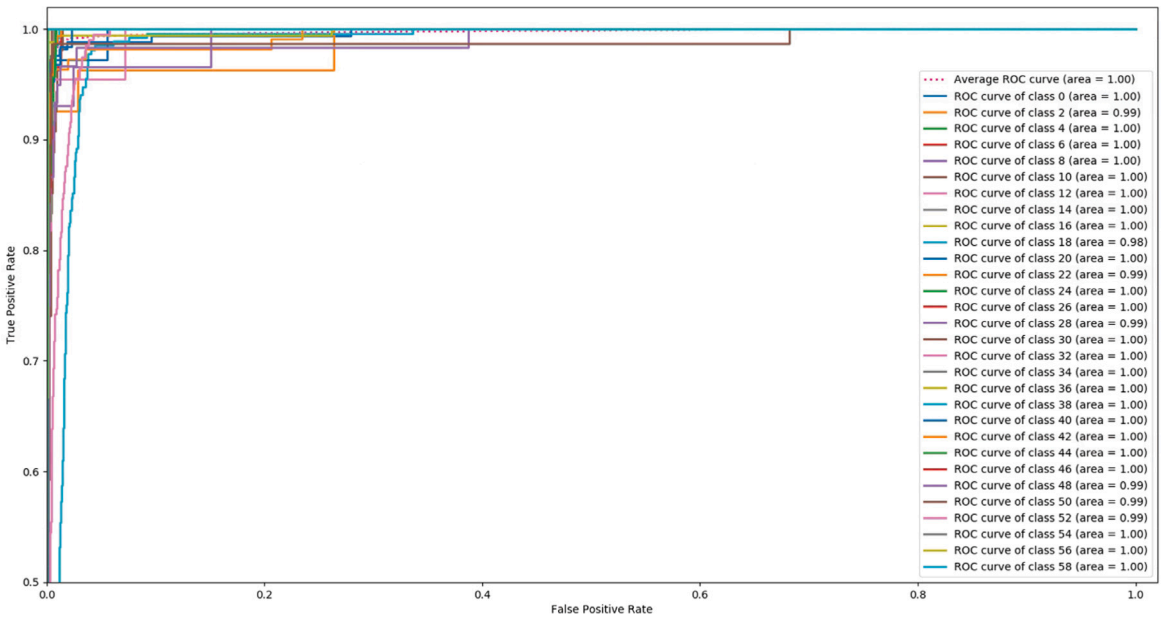


Fig. 15. ROC curves on the AI2018 dataset.

ICVT, and proves that inception transformer block and soft split token embedding are helpful for plant disease identification. From Table 9, it can be seen that the proposed ICVT can achieve the state-of-the-art performance 86.89% on the AI2018, 99.94% on the VillagePlant, 99.22% on the ibean, and 77.54% on the PlantDoc. Especially on the ibean, the ICVT method misclassified only one sample. These results validate the effectiveness of our ICVT.

5. Conclusion

Plant disease identification using digital images is a quite challenging. Therefore, it is very essential to well-timed discover of the plant disease and take the necessary action by farmers. In this paper, a novel deep learning-based network was proposed for automatic plant disease identification that was based on inception convolution and vision transformer. The ICVT architecture not

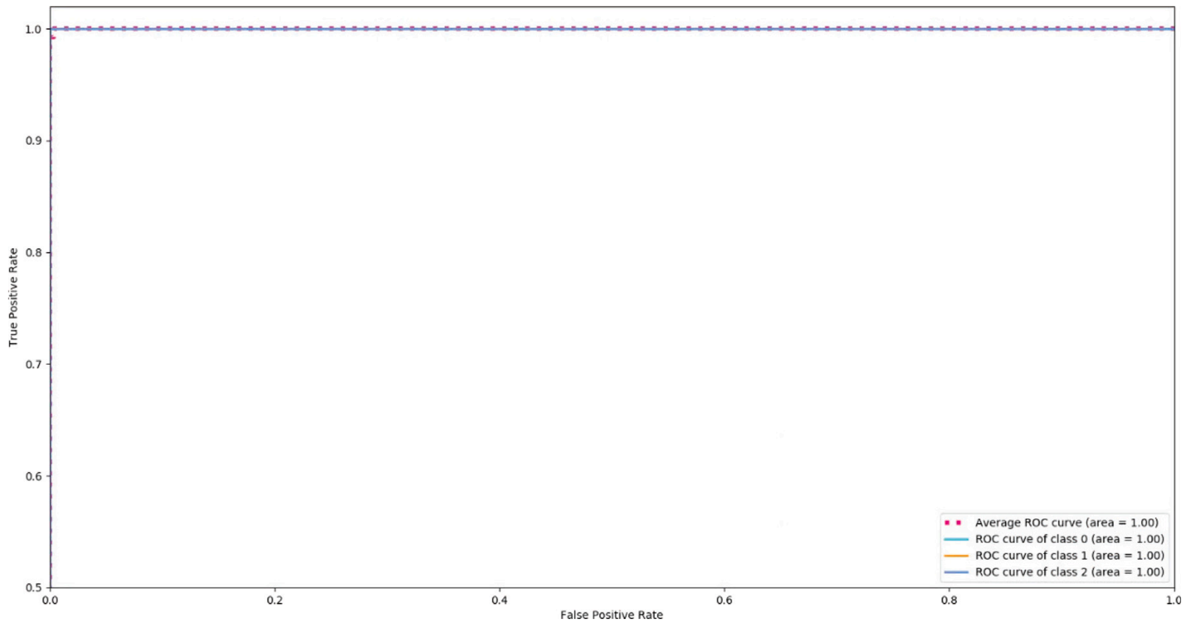


Fig. 16. ROC curves on the ibean dataset.

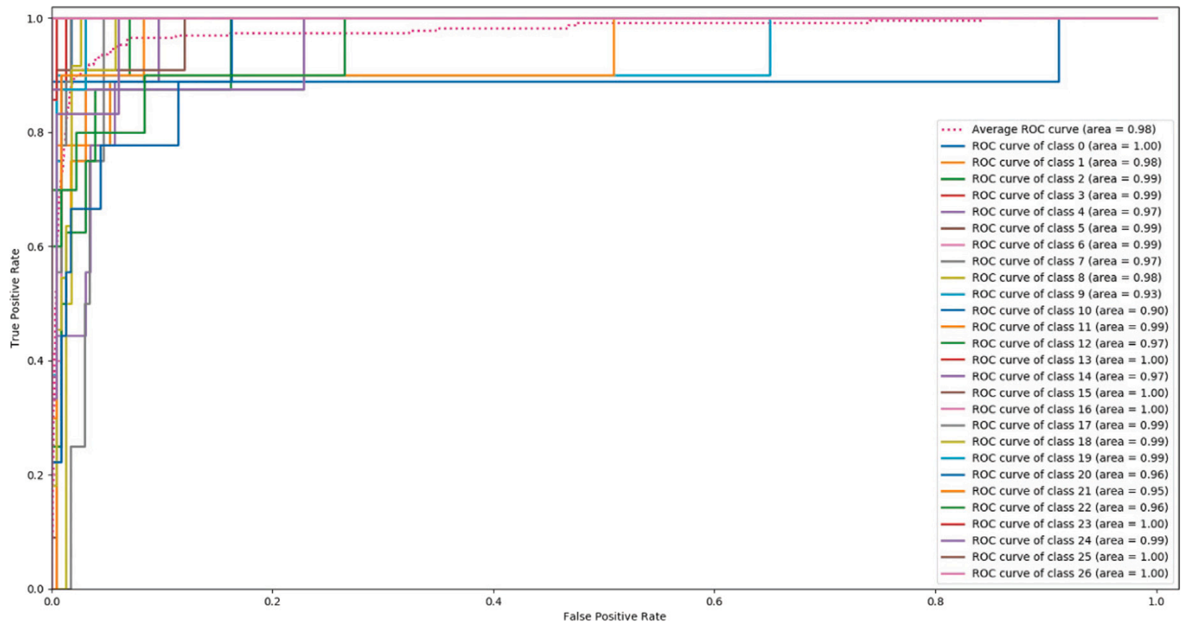


Fig. 17. ROC curves on the PlantDoc dataset.

only models the local spatial information, but also focuses on the learning of high-level information for plant disease identification. The proposed ICVT network achieves an average accuracy of 77.54%, 86.89%, 99.94%, and 99.22%, on the PlantDoc, AI2018, PlantVillage, ibean, respectively, and achieves the state-of-the-art performance.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sheng Yu reports administrative support, article publishing charges, equipment, drugs, or supplies, statistical analysis, travel, and writing assistance were provided by Shaoguan University. Sheng Yu reports a relationship with Shaoguan University that includes: employment. None

Data availability

The authors do not have permission to share data.

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